

Market Structure and the Hurst Coefficient

“The market has a definite structure,” said Tom randomly.

It is well understood that white noise has no correlation in time, and random walk (Brownian motion) noise has no correlation between increments. Brownian walks can be generated from a function where the spectral density is proportional to $1/F^\alpha$, where F is frequency, α is a power law, and $\alpha = 1$ generally signifies “long memory,” as I will describe in Chapter 8. Thus, $1/F$ noise is synonymous with long-range dependence. Integration of noise increases the exponent α by 2, whereas the inverse operation of differentiation decreases it by 2. Therefore, $1/F^\alpha$ noise cannot be obtained by the simple procedure of integration or of differentiation of such convenient signals. The widespread occurrence of signals exhibiting such behavior suggests that a generic mathematical explanation might exist. The ubiquity of $1/F^\alpha$ noise is one of the oldest puzzles of contemporary physics and science in general.

Figure 6.1 displays some time series and, in the same colors, their associated power spectra. Such time series arise in many natural systems. Power spectra are plotted in log-log coordinates, as is customary, because

$$\begin{aligned}\log(S(F)) &= \log(\text{constant}/F^\alpha) \\ &= \alpha \log(F) + \log(\text{constant})\end{aligned}$$

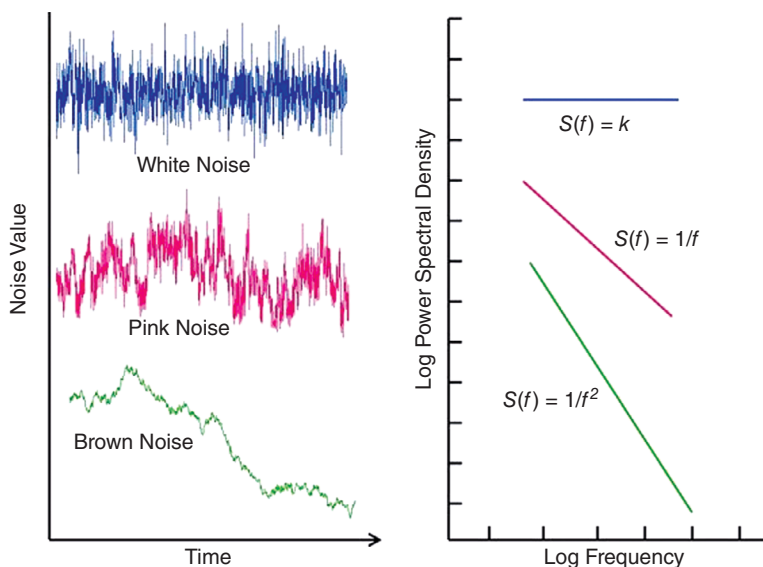


FIGURE 6.1 Time Series and Spectral Densities for Several Kinds of Noise

In other words, the logarithmic transform renders the $1/F^\alpha$ power spectrum a straight line whose slope, $-\alpha$, can be easily estimated. Clearly, for such natural systems, observed by humans, neither arbitrarily small nor arbitrarily large frequencies can be recorded. These models are simply relevant to physically plausible situations.

Noise in economic data is usually studied as long-range dependence or long memory. It has been shown repeatedly that the autocorrelation functions of economic time series, such as series of stock prices over days, weeks, or months, or the gross national product of various countries over years, do not decay exponentially as they would if the process generating the series were a simple autoregressive process. Instead, the autocorrelation functions of many economic time series reach a nonzero asymptote and remain there for the entire series, albeit often at a low value, indicating that economic events some distance in the past continue to have an influence on current prices. You only need to do an Internet search on “stock market spectral density” to find a number of academic papers, both measured and theoretical, regarding the frequency content of the market.

The Hurst coefficient is one way to attempt to get a handle on the slope of the power density of market data. The Hurst coefficient varies between

0 and 1, and is related to the α power coefficient as $H = 1 - \alpha/2$. The Hurst coefficient is more estimated than computed. I find the estimate using the fractal dimension is the most practical for shorter-term market data. The Hurst coefficient is related to the fractal dimension as $H = 2 - D$. I would like to make it perfectly clear that the Hurst coefficient or the fractal dimension has no direct practical application to trading not only because it is an estimate, but also because it has no predictive value. These computations only reflect the general structure of the market, and the answer you get is dependent on your assumptions. For example, the Hurst coefficient changes dramatically with the length of data used in making the estimate.

■ Fractal Dimension

To determine the fractal dimension of a generalized pattern, we cover the pattern with a number “ N ” of small objects of several various sizes “ S .” The relationship of the number of objects in two sets of sizes is:

$$\frac{N_2}{N_1} = \left(\frac{S_1}{S_2} \right)^D$$

$$D = \frac{\log(N_2/N_1)}{\log(S_1/S_2)}$$

As an example, we can start with a pattern that is a line segment 10 meters long. We chose the two small dimensions as $S_1 = 1$ meter and $S_2 = 0.1$ meter. Placing boxes along the line we can fit ten 1-meter boxes on the segment, therefore $N_1 = 10$. Similarly, we can fit one hundred 0.1-meter boxes on the same 10-meter line segment. Therefore, $N_2 = 100$. The fractal dimension of the line then computes to:

$$D = \frac{\log(100/10)}{\log(1/0.1)} = 1.0$$

As a second example, we will use the pattern as a square that is 10 meters on a side instead of a line segment. Retaining the same sizes of our small boxes as 1 meter and 0.1 meter on a side, respectively, we get $N_1 = 100$

and $N_2 = 10,000$. When the square is our pattern, the fractal dimension therefore computes to be:

$$D = \frac{\log\left(\frac{10000}{100}\right)}{\log\left(\frac{1}{0.1}\right)} = 2.0$$

A perfect square represents an idealized geometry not found in nature. Natural fractals, such as that of a seashore, lack the true regularity of an algorithmic structure but are self-similar in a statistical sense. Thus, in order to determine the fractal dimension of natural shapes, we must average the measured fractal dimension made over different scales.

We could measure the fractal dimension of prices by covering the curve with a series of small boxes. This is a burdensome task, but if we take into account that the price samples are uniformly spaced, we note that the box count is approximately the average slope of the curve. Therefore, we can estimate the box count as the highest price during an interval minus the lowest price during that interval, divided by the length of the interval itself. The equation for the number of boxes is then:

$$N_1 = \frac{\text{HighestPrice} - \text{LowestPrice}}{\text{Length}}$$

We compute the fractal dimension by computing N over two equal intervals to get the averaging over each interval. Interval 1 covers the period from 0 to T bars ago. Interval 2 covers the period from T to $2T$ bars ago. Therefore, $N_1 = (\text{Highest Price} - \text{Lowest Price})$ over the interval from 0 to T , divided by T . Similarly, $N_2 = (\text{Highest Price} - \text{Lowest Price})$ over the interval from T to $2T$, divided by T . We also define an $N_3 = (\text{Highest Price} - \text{Lowest Price})$ over the entire interval from 0 to $2T$, divided by $2T$. Since we are looking backwards in time, the slope computation of the fractal dimension is:

$$D = \frac{\text{Log}(N_1 + N_2) - \text{Log}(N_3)}{\text{Log}(2)}$$

The fractal dimension varies over the range from $D = 1$ to $D = 2$. Since the Hurst coefficient is related to the fractal dimension as $H = 2 - D$, so that it varies between 0 and 1. A value of 0.5 indicates a true random walk (a Brownian time series). In a random walk there is no correlation between any element and a future element. A coefficient between 0.5 and 1 indicates “persistent behavior,” that is, a long-term dependence. A Hurst coefficient

between 0 and 0.5 exists with time series with “antipersistent behavior”—in other words, “a countertrender” (a term that I detest).

■ Computing the Hurst Coefficient

The EasyLanguage code to compute the Hurst coefficient is given in Code Listing 6-1. The only user input is the length of data to be used. The number can be arbitrarily large if you have sufficient data. The results are critically dependent on the input data length selected. After declaring variables, the coefficients of a 20-bar SuperSmoother filter are computed. The computations of $N1$, $N2$, and $N3$ are as described in the previous section. The fractal dimension is then converted to the Hurst coefficient, which is subsequently smoothed in the SuperSmoother filter.

Code Listing 6-1. EasyLanguage Code to
Compute the Hurst Coefficient

```
{
    Hurst Coefficient
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}

Inputs:
    Length(30); {Length must be an even number}

Vars:
    a1(0),
    b1(0),
    c1(0),
    c2(0),
    c3(0),
    count(0),
    N1(0),
    N2(0),
    N3(0),
    HH(0),
    LL(0),
    Dimen(0),
    Hurst(0),
    SmoothHurst(0);
```

(Continued)

```
//Smooth with a Super Smoother Filter from equation 3-3
a1 = expvalue(-1.414*3.14159 / 20);
b1 = 2*a1*Cosine(1.414*180 / 20);
c2 = b1;
c3 = -a1*a1;
c1 = 1 - c2 - c3;

N3 = (Highest(Close, Length) - Lowest(Close, Length)) /
Length;
HH = Close;
LL = Close;
For count = 0 to Length / 2 - 1 begin
    If Close[count] > HH then HH = Close[count];
    If Close[count] < LL then LL = Close[count];
End;
N1 = (HH - LL) / (Length / 2);
HH = Close[Length / 2];
LL = Close[Length / 2];
For count = Length / 2 to Length - 1 begin
    If Close[count] > HH then HH = Close[count];
    If Close[count] < LL then LL = Close[count];
End;
N2 = (HH - LL) / (Length / 2);
If N1 > 0 and N2 > 0 and N3 > 0 then Dimen = .5*((Log(N1
+ N2) - Log(N3)) / Log(2) + Dimen[1]);
Hurst = 2 - Dimen;
SmoothHurst = c1*(Hurst + Hurst[1]) / 2 + c2*SmoothHurst[1]
+ c3*SmoothHurst[2];

Plot1(SmoothHurst);
```

■ The Hurst Coefficient in Action

The fractal dimension indicator is applied to approximately one year of data of Dollar General (symbol DG) in Figure 6.2 using an input length of 30. With this setting, the Hurst coefficient swings between showing persistence and antipersistence. However, when the input length is increased to 200, as shown in Figure 6.3, the uptrend is reflected as the Hurst coefficient rising to be above 0.5.

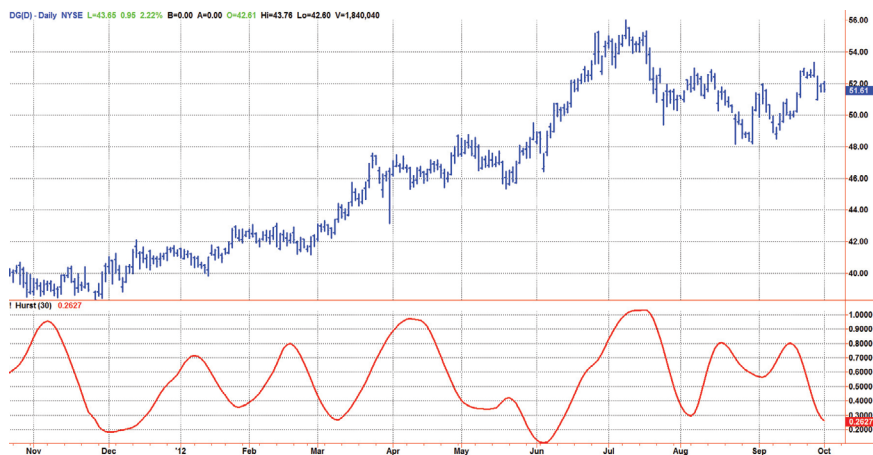


FIGURE 6.2 Using an Input Length of 30, the Hurst Coefficient Swings between Showing Persistence and Antipersistence

It is perhaps instructive to describe action of the Hurst coefficient from another theoretical perspective. In Figure 6.4, the indicator is applied to a theoretical sine wave whose period is continuously increasing from left to right from a period of 10 bars to 40 bars. A period of 20 bars is approximately in the horizontal center of the chart. In this case, I have given the Hurst coefficient indicator an input of 20. The periodicity in the persistent case is due to the peaks and valleys of the time waveform tending to “fill the

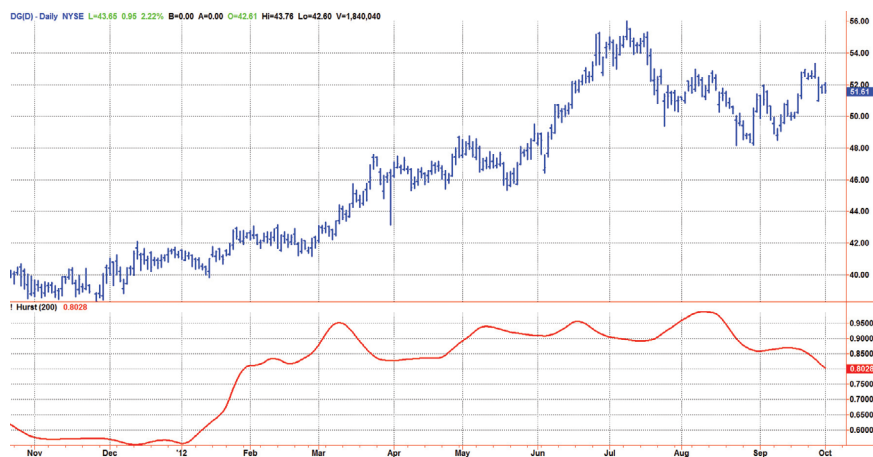


FIGURE 6.3 The Longer-Term Trend is Indicated by a Hurst Coefficient Greater than 0.5 When an Input Length of 200 Is Used

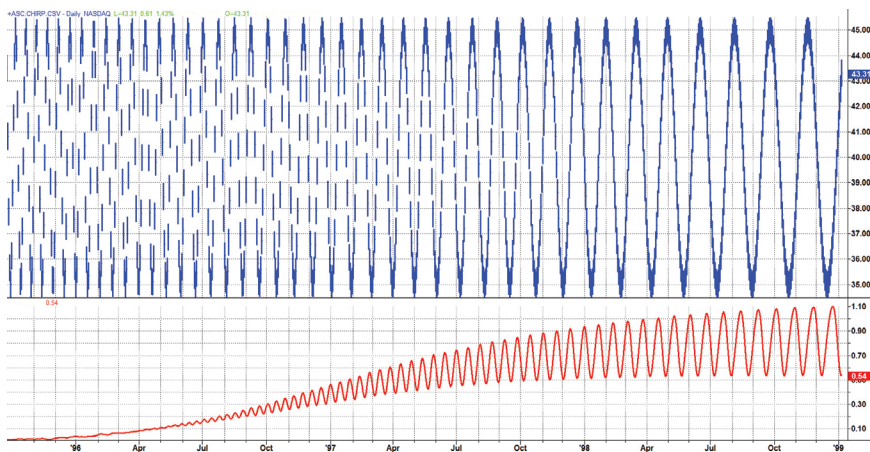


FIGURE 6.4 Cycle Periods Shorter than the Hurst Coefficient Input Parameter Display as Antipersistent; Longer Cycle Periods Display as Persistent

boxes” of the fractal dimension. From this perspective, the Hurst coefficient can be used to define trend modes (> 0.5) and cycle modes (< 0.5) in the market relative to the selected input length.

■ Drunkard's Walk Hypothesis for Market Structure

The efficient markets model statement that the price fully reflects available information has been assumed to imply that successive price changes are independent of each other. In addition, it has usually been assumed that successive changes are identically distributed. Together, these two hypotheses constitute the random walk model. This model says that the conditional and marginal probability distributions of an independent random variable are identical. In addition, it says that the probability density function must be the same for all time. This model is clearly flawed. If the mean return is constant over time then the return is independent of any information available at a given time.

I assume that there are an adequate number of traders involved in making the market that a statistical analysis involving a random walk is appropriate. There must be several constraints to such a random walk. The first constraint is that the prices are constrained to one dimension—they can only go up or down. The second constraint is that time must progress monotonically.

I propose a philosophical basis of market action from extensive work using constrained random walks in the physical sciences.¹ The expression of such a random walk is that of a drunkard moving on a one-dimensional array of regularly spaced points. At regular intervals the drunkard flips a coin and makes one step to the right or left, depending on the outcome of the coin toss. At the end of n steps, he can be at any one of $2n + 1$ sites, and the probability that he is at any site can be calculated. Let the distance between the points on the lattice be ΔL , and let the time between successive steps be ΔT . If ΔL and ΔT are allowed to shrink to zero in such a way that $(\Delta L)^2 / \Delta T$ remains constant to the diffusion constant D , then the equation governing the distribution of the displacement of the random walker from his starting point is

$$\frac{\delta P}{\delta t} = D \frac{\delta^2 P}{\delta x^2}$$

This rather famous partial differential equation is called the *diffusion equation*. The function $P(x, t)$ can be interpreted in two ways. It can be taken to express either the probability density or the concentration of diffusing matter at position x at time t . Following the latter interpretation, it can, for example, describe the way heat flows up the stem of a silver spoon when placed in a hot cup of coffee.

To better understand the theory of diffusion, imagine the way a smoke plume leaves a smokestack. Think about how the smoke rises as comparable to how a trend carries itself through the market. A gentle breeze determines the angle to which the smoke, or trend, is bent. The widening of the smoke plume represents the probability density of the smoke particles as a function of distance from the smokestack. This widening is analogous to the decreased accuracy of the prediction of future trend prices further into the future.

The formulation of the drunkard's walk has no property that can be regarded as the analog of momentum. A more realistic model of the motion of a physical object needs to take into account some form of memory—we need to know where the object came from and the likelihood that it will continue to move in the same direction. The simplest modification of the random walk is to allow the coin toss to determine the persistence of motion. In other words, with probability p the drunkard makes his next step in the same direction as the last one, and with probability $1 - p$ he makes a move in the opposite direction. The ordinary drunkard's walk occurs when $p = 1/2$ because either move is equally likely. The interesting feature of the modified drunkard's walk is that as the distance between the point and the

time between steps decreases, one no longer obtains the diffusion equation, but rather the following equation:

$$\frac{\delta^2 P}{\delta t^2} + \frac{1}{T} \frac{\delta P}{\delta t} = c^2 \frac{\delta^2 P}{\delta x^2}$$

in one dimension, where T and c^2 are constants. This is another famous partial differential equation called the *telegrapher's equation*. This equation expresses the idea that diffusion occurs in restricted regions, such that $x^2 < c^2 t^2$. That is, the position must be less than the velocity of propagation (c) multiplied by time (t). More important, the telegrapher's equation describes the harmonic motion of $P(x, t)$ just as surely as it describes the electric wave traveling down a pair of wires.

Harmonic motion is ubiquitous. It is the natural response to a disturbance on any scale ranging from the atomic to the galactic. You can demonstrate the effect to yourself by holding a ruler over the edge of a table, bending the ruler down, and then releasing it. The resulting vibration is harmonic motion. Alternatively, you can stretch a rubber band between your fingers, pull the band to one side, and then release it. The oscillations of the rubber band also constitute harmonic motion. Since there are plenty of opportunities for market disturbances, it is only a small stretch to extend the solution to the drunkard's walk problem from physical phenomena and use it to describe the action of the market.

The drunkard's walk solution can describe two market conditions. The first condition, where the probability is evenly divided between stepping to the right or the left, results in the *trend mode*, described by the diffusion equation. The second condition, where the probability of motion direction is skewed, results in the *cycle mode*, described by the telegrapher's equation. The difference between the two conditions can be as simple as the question that the majority of traders are constantly asking themselves. If the question is "I wonder if the market will go up or down?," then the probability of market movement is about 50–50, establishing the conditions for a trend mode. However, if the question is posed as "Will the trend continue?," then the conditions are such that the telegrapher's equation applies. As a result, the cycle mode of the market can be established.

The telegrapher's equation solution also describes the meandering of a river. Viewed as an aerial photograph, every river in the world meanders. This meandering is not due to a lack of homogeneity in the soil, but to the conservation of energy. (You can appreciate that soil homogeneity is not

a factor because other streams, such as ocean currents, also meander in a nearly homogeneous medium.) Ocean currents are not nearly as visible as rivers and are therefore not as familiar to most of us. Every meander in a river is independent of other meanders, and thus all are completely random. If we were to look at all the meanders as an ensemble, overlaying one on top of the other like a multiple-exposure photograph, the meander randomness would also become apparent. The composite envelope of the river paths would be about the same as the cross-section of the smoke plume. However, if we are in a given meander, we are virtually certain of the general path of the river for a short distance downstream. The result is that the river can be described as having a short-term coherency but is random over the longer span.

River meanders are the kind of cycles we have in the market. We can measure and use these short-term cycles to our advantage if we realize they can come and go in the longer term.

We can extend our analogy to understand when short-term cycles occur. Rivers meander in an attempt to maintain a constant slope on their way to the ocean. If the slope is too severe, the meanders have the same effect as a skier who weaves back and forth across the slope to slow his descent. The flow of a river physically adjusts itself for the purpose of energy conservation. If the water speeds up, the width of the river decreases to yield a constant flow volume. The faster flow contains more kinetic energy, and the river attempts to slow it down by changing direction. At the same time, the river direction cannot change abruptly because of the momentum of the flow of water. Meandering results. Thus, meanders cause the river to take the path of least resistance in the sense of energy conservation. We should think of markets the same way. Time must progress as surely as the river must flow to the ocean. Overbought and oversold conditions result from an attempt to conserve the “energy” of the market. This “energy” arises from the fear and greed of traders.

Again, it may be useful to test the principle of conservation of energy for yourself. Tear a strip about 1 inch wide along the side of a standard sheet of paper about 11 inches long. Grasp each end of this strip between the thumb and forefinger of each hand. Now move your hands toward each other. Your compression is putting energy into this strip, and its natural response can take one of four modes. These modes are determined by the boundary conditions that you forced. If both hands are pointing up, the response is a single upward arc, approximating one alternation of a sine wave. If both hands are pointing down, the response is a downward arc. If

one hand is pointing up and the other pointing down, the strip response to the energy input is approximately a full sine wave. The four lowest modes are the natural responses following the principle of conservation of energy. You can introduce additional bends in the strip, but a minor jiggling will cause the paper to snap to one of the four lowest modes, with the exact mode depending on the boundary conditions that you impose. The two full sine wave modes are approximately the second harmonic of the two single-alternation modes.

The market only has a single dominant cycle most of the time, as shown by the spectrum measurements in Chapters 8 through 10. When multiple cycles are simultaneously present, they are generally harmonically related. This is not to say that nonharmonic simultaneous cycles cannot exist, just that they are rare enough to be discounted in simplified models of market action. The general observation of a single dominant cycle tends to support the notion that the natural response to a disturbance is monotonic harmonic motion.

It is true that if you are a hammer, the rest of the world looks like a nail. We must take care to recognize that all market action is not strictly described by cycles alone and that cycle tools are not always appropriate. A more complete model of the market can be achieved by recognizing the fact that there are times when the solution to the telegrapher's equation prevails and times when the solution to the diffusion equation applies. We can therefore divide the market action into a cycle mode and a trend mode. By having only two modes in our market model we can switch our trading strategy back and forth between them, using the more appropriate tool according to our situation. Since our digital signal processing tools analyze cycles, we can establish that a trend mode is more appropriate at any given time due to the failure of a cycle mode.

■ Key Points to Remember

1. The markets can be characterized as noise having varying degrees of long-term memory.
2. The Hurst coefficient is related to the power spectral density of the noise and to the fractal dimension. The use of these terms is interchangeable.
3. The Hurst coefficient is more estimated than computed.
4. The Hurst coefficient has no predictive value and therefore has no direct usefulness in trading.

5. The appearance of the Hurst coefficient is critically dependent on the length of data used to compute it.
6. A Hurst coefficient less than 0.5 indicates a cycle mode.
7. A Hurst coefficient greater than 0.5 indicates a trend mode.
8. A drunkard's walk model can also be used to form a philosophical basis of market activity.

Note

1. G. H. Weiss and R. J. Rubin, "Random Walks: Theory and Selected Applications," *Advances in Chemical Physics* 52 (1982): 363–505.